

KRCM / CBHO

Complete Working Record

Public critique brief - focused on Tests 52-54

Kinetic Recoil Cosmological Model / Cosmic Black Hole Object

Compiled 22 September 2026

Working-status note. This is a speculative, non-peer-reviewed research record. PASS means a stated numerical or mathematical gate passed inside the declared model; it does not mean the physical hypothesis is established. Reduced, proxy, local, and conditional results are labeled as such.

Purpose

This is a short critique packet, not a claim of established physics. It isolates the current strongest calculation so another person can attack the equations, boundary conditions, free-data choice, characteristic analysis, and nonlinear evolution without reading the entire project history.

Model

$$S = \text{integral} \sqrt{-g} [F R - 1/2(dh)^2 - 1/2(d\phi)^2 - V + f G] d^4x$$

$$F = 1 + 0.08 h^2 + 0.06 \phi^2; f = 0.82 h + 0.68 \phi$$

$$V = 0.300 + 0.25(h^2-1)^2 + 0.20(\phi^2-1)^2 + 0.09(h-\phi)^2 - 0.070h - 0.055\phi$$

Current simulations use dimensionless model units unless explicitly stated. Parent minimum: $h=-0.9630260602$, $\phi=-0.9635971213$, $V=0.4227485835$, $F=1.1299047001$. Daughter minimum: $h=1.0332773412$, $\phi=1.0327977708$, $V=0.1729002789$.

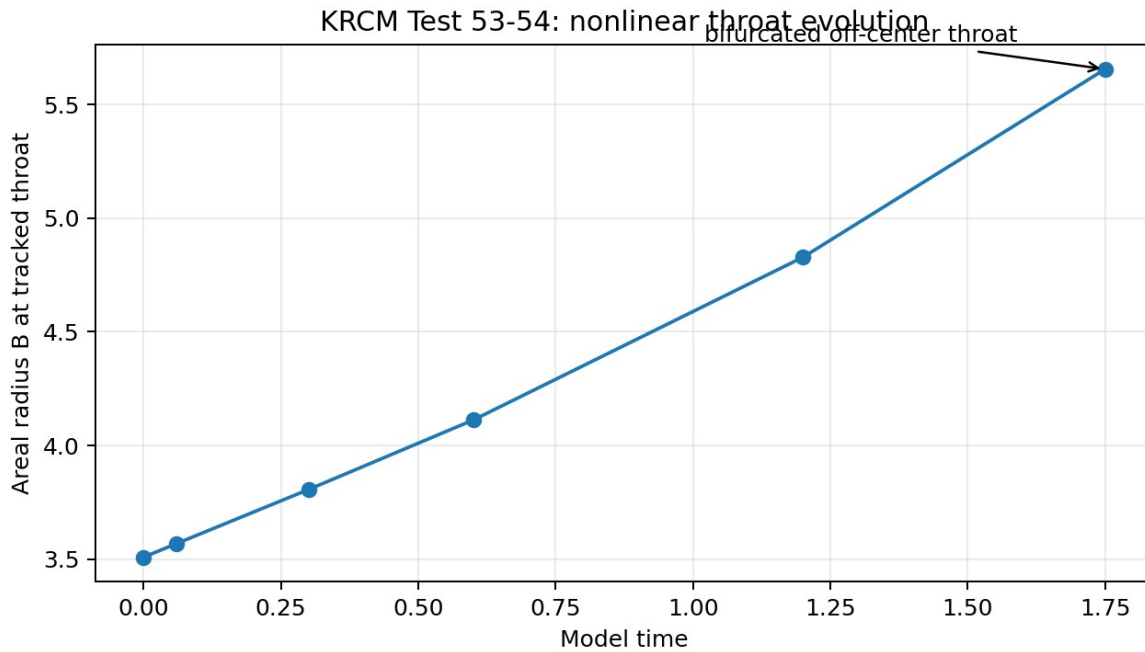
Strongest current result: Test 52

$$H_{\text{parent}} = \sqrt{V_{\text{parent}}/(6 F_{\text{parent}})} = 0.2497149524$$

- The previous static-parent target W approximately 0 was replaced by the correct expanding parent branch $W, U \rightarrow H_{\text{parent}}$.
- A monotone single-throat slice was found with $B'(l) \geq 0$ everywhere.
- Hamiltonian residual $\leq 1.11e-16$ and momentum residual $\leq 1.39e-17$ in the exact initial-data construction.
- At $dr=0.04, 0.02, 0.01$ the physical principal symbol has four real propagating roots across roughly 8,000 radial points per run.
- Minimum principal r_{cond} is about 0.0591; maximum physical eigenvector condition about 5.92.

Nonlinear follow-up

- Test 53: to $t=0.06$ the throat radius grows $\sim 1.704\%$, B_{xx} at the throat stays positive, F stays positive, final physical roots stay real, and absolute constraints converge at nearly second order.
- Test 54: extending the evolution causes the center to change from an areal-radius minimum to a maximum, while symmetric off-center throats form. At $t=1.75$ the positive-side throat is near $x=0.47$ with $B \sim 5.65575$ and $B_t \sim 1.6212$.
- The late-time bifurcation is only a conditional result because its exact onset and late-time hyperbolicity still need finer, independent-gauge convergence.



What I want critics to attack

1. Is the variation and boundary treatment of the scalar-dependent $F R + f G$ action correct?
2. Is the Test-52 free-data construction a legitimate initial-data choice, or does it hide an unphysical tuning?
3. Is the local physical principal-symbol calculation sufficient to support the stated hyperbolicity gate?
4. Could the Test-54 double-throat bifurcation be a synchronous-gauge or finite-difference artifact?
5. What invariant diagnostic or independent formulation would most efficiently kill or confirm this branch?

Important limitations

- Not peer reviewed.
- No derivation of F , f , V from a UV-complete fundamental theory.
- No dimensional calibration to real cosmological scales yet.
- Current CMB/BBN/late-time-acceleration calculations are not yet rerun on the Test-52 global branch.
- Long-time production-grade numerical relativity remains unfinished.

Reddit-ready disclosure

AI disclosure: LLM tools were used for algebra/code organization, numerical test design, debugging, and report generation. The equations, source files, and raw numerical outputs should be treated as the object of critique and checked independently.

Suggested title: "Here is a hypothesis: A curvature-coupled two-scalar Gauss-Bonnet model can support a persistent daughter-region throat"

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